

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 HT	(a)	(i)		CaCO ₃ (1)+ CO ₂ (1)		2		2		
		(ii)		Calcium sulfate is present in permanent hard water (1) not broken down by boiling (1)	2			2		
	(b)	(i)		forms a lime scale (1) calcium carbonate precipitates (1)	2			2		
		(ii)		uses (concentrated) NaCl in resin columns (1) (two) Na ions <u>exchanged</u> for calcium ions. (1)	2			2		
	(c)			Needs to be replaced after time	1					
	(d)			causes {blocked pipes/limescale/furring} inefficient boilers (1)	2					
				Question 6 total	9	2	0	11	0	0